The Topology Confound: Why Multi-Agent Systems Cannot Distinguish Cycles from Density

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Abstract

Multi-agent AI systems increasingly rely on communication topologies to coordinate behavior, yet the field lacks principled methods for reasoning about which topological properties drive system dynamics. We prove a fundamental methodological confound: for connected graphs with n vertices and m edges, the cycle rank $\beta_1 = m - n + 1$, making cycle structure and edge density algebraically indistinguishable in every published topology comparison. At least seven major multi-agent systems enforce DAG communication topologies based on confounded evidence. We escape this identity via directed simple cycle count $\kappa(D)$, which decorrelates from density, and present the first controlled experiment: 240 island model runs at constant density show a significant effect of cycle structure on population diversity ($p < 10^{-7}$, $\eta^2 = 0.17$). For AI safety, the implication is immediate: claims about the superiority of acyclic agent coordination are unfounded, and the field is deploying topologies whose behavioral consequences are not understood.

1 Introduction

The topology of communication in multi-agent systems—which agents can talk to which—shapes collective behavior in ways that are poorly understood and almost never controlled. In island model evolutionary algorithms, ring topologies maintain diversity longer than complete graphs (**?**); sparse topologies slow premature convergence (**?**). In LLM-based multi-agent systems, seven recent architectures enforce directed acyclic graph (DAG) communication (**???????**), claiming that acyclic structure improves performance.

From a safety perspective, this matters. Multi-agent systems are being deployed in settings where predictable, controllable behavior is essential: code generation pipelines, automated reasoning chains, and autonomous agent swarms. If the field's understanding of *why* certain topologies work rests on confounded evidence, then topology choices in deployed systems are effectively uncontrolled—designers are tuning a knob without knowing what it does.

We show that this is precisely the situation. The cycle rank β_1 of a connected graph with n vertices and m edges is $m - n + 1$: cycle structure and density are the *same variable*. Every comparison of "cyclic vs. acyclic" topologies simultaneously varies how many communication channels exist. The DAG hegemony may reflect nothing more than the cost savings of sparser communication.

To break this confound, we move to directed graphs, where the simple directed cycle count $\kappa(D)$ depends on arc arrangement, not just (n, m). We construct eight graph families at constant density and run 240 controlled experiments. The result: cycle structure has an independent, significant effect on diversity dynamics—an effect invisible to every prior experimental design.

2 The Density–Cycle Confound

Theorem 1 (Cycle rank–density identity)**.** *Let $G = (V, E)$ be a connected graph with $|V| = n$ and $|E| = m$. Then $\beta_1(G) = m - n + 1$.*

Proof. A spanning tree of G uses $n - 1$ edges. Each remaining edge closes exactly one independent cycle. The result follows from the Euler characteristic $\chi = n - m = 1 - \beta_1$ for connected 1-dimensional simplicial complexes (**?**). □

Corollary 2. *For connected graphs with fixed n, β_1 and m differ by the constant $n-1$. Pearson correlation is affine-invariant, so $\rho(\beta_1, Y) = \rho(m, Y)$ for any outcome variable Y.*

The methodological consequence: *any experiment comparing connected topologies on a fixed node count cannot distinguish "cycles affect performance" from "edge count affects performance."* This is not a subtle statistical concern—it is a mathematical identity.

Scope of the confound. Table **??** surveys seven systems that enforce DAG communication. In every case, the comparison that motivates the DAG

Table 1: Systems enforcing the DAG constraint. All vary β_1 and density simultaneously.

System	Confound
GoAgent	DAG vs. dense baseline varies β_1 and m
Graph-GRPO	Complete graph = max density *and* max β_1
OFA-MAS	Pruning reduces m and β_1 in lock-step
G-Designer	No density-controlled baseline
AgentConductor	Cyclic alternatives never tested
Dochkina et al.	Sparsity and acyclicity confounded in 25K runs
HERA	Tracks cycle count but not at constant density

constraint simultaneously varies cycle structure and density. GoAgent's CIB objective (**?**) compares DAGs to dense baselines; Graph-GRPO (**?**) learns DAGs and evaluates against complete graphs (maximum density *and* maximum β_1); Dochkina et al. (**?**), the most rigorous study (25,000+ runs), confounds sparsity and acyclicity throughout.

The safety implication. If we cannot distinguish "fewer cycles help" from "fewer edges help," then the DAG constraint in deployed systems may be discarding useful feedback pathways (cycles that enable error correction and redundancy) while the actual benefit comes from reduced communication overhead. A system designed for safety through acyclicity may be safe only because it is sparse—and adding edges for robustness would reintroduce cycles, with unknown behavioral consequences.

3 The Directed Escape

Theorem **??** applies to the homological cycle rank. For directed graphs, the *simple directed cycle count* $\kappa(D)$—the number of elementary directed cycles—is combinatorial, not homological. It depends on the specific arc arrangement, not just on (n, m).

We constructed eight directed graph families, all with $n = 8$ vertices and $m = 16$ arcs (density $m/n = 2.0$ held constant). The families span κ from 0 (pure DAGs) to 47 (ring with skip-2 connections). Among strongly connected members, κ ranges from 10 to 47—a 4.7× variation at strictly constant density. Table **??** summarizes.

Table 2: Eight directed graph families at constant density ($n = 8$, $m = 16$).

Family	κ	SC
`dag-layer`	0	No
`dag-wide`	0	No
`lowcyc-1`	3	No
`bidir-ring`	10	Yes
`two-cliques`	14	No
`mesh-cyclic`	20	Yes
`dense-triangles`	29	Yes
`ring-skip2`	47	Yes

4 Experiment: Constant-Density GA

Setup. Each of the 8 graph families serves as an island model GA migration topology. Each vertex hosts a subpopulation of 50 individuals (400 total). Fitness function: OneMax on 100-bit strings ($f(\mathbf{x}) = L^{-1} \sum_k x_k$). Selection: tournament ($k = 3$). Crossover: uniform ($p = 0.5$/locus). Mutation: bit-flip at rate $1/L$. Every 10 generations, the 5 fittest individuals migrate along outgoing arcs, replacing the 5 weakest. All families share density $16/8 = 2.0$. Each topology runs with 30 random seeds (240 runs total, 500 generations each).

Diversity metric. Mean pairwise normalized Hamming distance across all 400 individuals, estimated from 20 randomly sampled pairs.

Statistical analysis. One-way ANOVA with topology as factor; Kruskal–Wallis for non-parametric confirmation; Pearson r between κ and metrics at generations 30, 60, and 100. By construction, $r(\text{density}, Y) = 0$ identically for all outcomes Y.

4.1 Results

Main effect. One-way ANOVA at generation 30: $F(7, 232) = 6.90$, $p = 1.85 \times 10^{-7}$. Kruskal–Wallis corroborates: $H = 38.53$, $p = 2.40 \times 10^{-6}$. Effect sizes: $\eta^2 = 0.17$ for diversity, $\eta^2 = 0.24$ for fitness—medium-to-large by Cohen's conventions.

Cycle count predicts diversity. Pearson $r = -0.681$ between κ and mean diversity at generation 30 ($p = 0.063$ on 6 df; marginal on 8 topology-level points, but corroborated by the 240-run ANOVA). Higher cycle count correlates with faster diversity loss: more feedback loops accelerate the spread of dominant alleles across islands.

Transience. The effect peaks at generations 20–50 and decays by generation 100 ($|r| < 0.10$, $\eta^2 = 0.002$). OneMax is unimodal, so all topologies converge eventually. κ affects the *rate* of convergence, not the outcome. DAG families preserve diversity longest; `ring-skip2` ($\kappa = 47$) converges fastest.

Interpretation. Even on the simplest possible fitness landscape, directed cycle structure independently explains 17% of diversity variance and 24% of fitness variance during the critical early phase. This is the first demonstration that cycles affect dynamics *at constant density*.

5 Implications for AI Safety

Uncontrolled topology is uncontrolled behavior. Our theorem shows that the field's evidence base for topology design is confounded. Systems deployed with DAG topologies may behave predictably not because they lack cycles, but because they are sparse. If a system is scaled up by adding communication channels (e.g., for fault tolerance or richer coordination), cycles will inevitably appear—and their behavioral effects, as our experiment demonstrates, are real and significant.

Diversity as a safety mechanism. In evolutionary computation, diversity maintenance prevents premature convergence to suboptimal solutions. In multi-agent AI, the analog is *behavioral diversity*: the system's ability to explore multiple solution strategies rather than collapsing to a single mode. High-κ topologies accelerate diversity loss—equivalent to the system "making up its mind" faster. Whether this is desirable depends on the task: for safety-critical applications where premature consensus is dangerous (e.g., multi-agent deliberation for high-stakes decisions), understanding the topology–diversity relationship is essential.

The landscape matters. A pilot study on NK landscapes (**?**) (rugged, multi-peaked fitness functions) shows that topology effects scale dramatically with problem difficulty: η^2 increases from 0.05 ($K = 0$, smooth) to 0.69 ($K = 6$, rugged)—a 14× amplification. Real-world problems facing deployed AI systems are closer to $K = 6$ than to OneMax. The moderate effects we observe on a toy problem are likely lower bounds on what matters in practice.

Toward principled topology design. We propose that AI safety frameworks for multi-agent systems must include topology as a first-class design variable, with three requirements: (1) density and cycle structure must be controlled independently; (2) diversity dynamics, not just task performance, must be monitored; (3) topology choices must be justified against constant-density baselines, not confounded comparisons.

6 Related Work

HERA (**?**) is the first system to track cycle count during topology evolution, converging to "compact chains with strategically preserved cycles." Puppeteer (**?**) discovers cyclic structures under optimization pressure, outperforming DAG-constrained baselines. Both provide evidence that cycles are functionally useful, but neither controls for density. Our theorem explains why their results are promising but not conclusive, and our experimental design provides the controlled condition they lack.

In evolutionary computation, the topology–diversity relationship is well-established (**??**), but all prior work uses undirected graphs and therefore falls under the confound. The directed cycle count $\kappa(D)$ as an experimental variable is, to our knowledge, novel.

7 Conclusion

For connected graphs, cycle rank equals edge count minus a constant. This identity means that every published comparison of cyclic vs. acyclic multi-agent topologies is confounded. We escape the confound via directed simple cycle count, which varies independently of density, and demonstrate a significant independent effect ($p < 10^{-7}$) across 240 controlled runs. Preliminary NK landscape results suggest 14× amplification on harder problems.

For AI safety, the message is: the field is building multi-agent systems with uncontrolled topology. The DAG hegemony rests on confounded evidence. Before we can reason about the safety properties of agent communication structures, we must first ensure we are measuring the right thing. This paper provides the theorem that exposes the problem and the experimental methodology that begins to solve it.

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